

Software Requirements Specification (SRS)

Preface

This document defines the software requirements for the Asset & Resource Allocation Management System. It is intended for students, instructors, project reviewers, developers, and testers.

Version History

Version	Date	Description
1.0	Initial	First version of the Software Requirements Specification created for academic submission.

Rationale: This document has been created as part of the Requirement Engineering process to formally specify system requirements.

1. Introduction

1.1 Need for the System

Organizations often rely on manual or fragmented methods to manage assets and resources. This results in inefficient utilization, lack of transparency, delayed approvals, and difficulty in tracking maintenance and allocation history.

1.2 System Overview

The Asset & Resource Allocation Management System is a centralized web-based application that manages asset tracking, allocation requests, approval workflows, and maintenance activities.

1.3 System Functions (Brief)

User and role management
Asset management and tracking
Resource request and approval workflow
Maintenance and issue tracking
Reporting and notifications

1.4 Interaction with Other Systems

The system is designed to work as a standalone application but may integrate with authentication or inventory systems in the future.

1.5 Business and Strategic Objectives

Improve asset utilization
Increase transparency and accountability
Reduce operational delays
Support informed decision-making

2. Glossary

Domain specific definitions

Term	Definition
Asset	A physical resource such as computers, equipment, or rooms
Allocation	Assignment of an asset to a user or office
Higher Authority	Senior authority responsible for high-level approvals
Audit Log	Record of system activities for accountability

3. User Requirements Definition

This section describes the services provided to system users. Requirements are expressed in natural language and are understandable to non-technical stakeholders.

3.1 User

- The system shall allow users to submit asset or resource requests.
- The system shall allow users to view request status.
- The system shall notify users of approval or rejection decisions.
- The system shall allow users to report asset-related issues.

3.2 Staff

- The system shall allow staff to add, update, and remove assets.
- The system shall allow staff to allocate assets to users or offices.
- The system shall allow staff to record maintenance details.

3.3 Office Admin

- The system shall allow office admins to approve or reject requests.
- The system shall allow office admins to escalate requests to the Higher Authority.

3.4 Non-Functional Requirements (User Level)

- The system shall be easy to use.
- The system shall provide secure access to all users.

4. System Architecture

The system follows a web-based client–server architecture:

Presentation Layer: Web interface for all actors.

Application Layer: Business logic, workflows, and access control.

Data Layer: Database for users, assets, requests, and audit logs.

Reusable components include authentication and role-based authorization modules.

5. System Requirements Specification

5.1 Functional Requirements

5.1.1 User Management

- The system shall allow users to log in and log out securely.
- The system shall allow users to reset passwords using email verification.
- The system shall support role-based access control.
- The system shall allow the System Admin to manage users and roles.

5.1.2 Asset Management

- The system shall allow staff to add, update, and delete assets.
- The system shall track asset status and maintenance history.
- The system shall display office-wise asset availability in real time.

5.1.3 Resource Allocation

- The system shall allow users to request asset allocation for specific periods.
- The system shall support approval, rejection, and escalation workflows.
- The system shall prevent double booking of assets.

5.1.4 Reporting and Notifications

- The system shall generate usage and activity reports.
- The system shall send email and in-app notifications for key events.

5.2 Non-Functional Requirements

Performance: System response time shall not exceed 3 seconds under normal load.

Security: Role-based authorization and audit logging shall be enforced.

Availability: The system shall be available during organizational working hours.

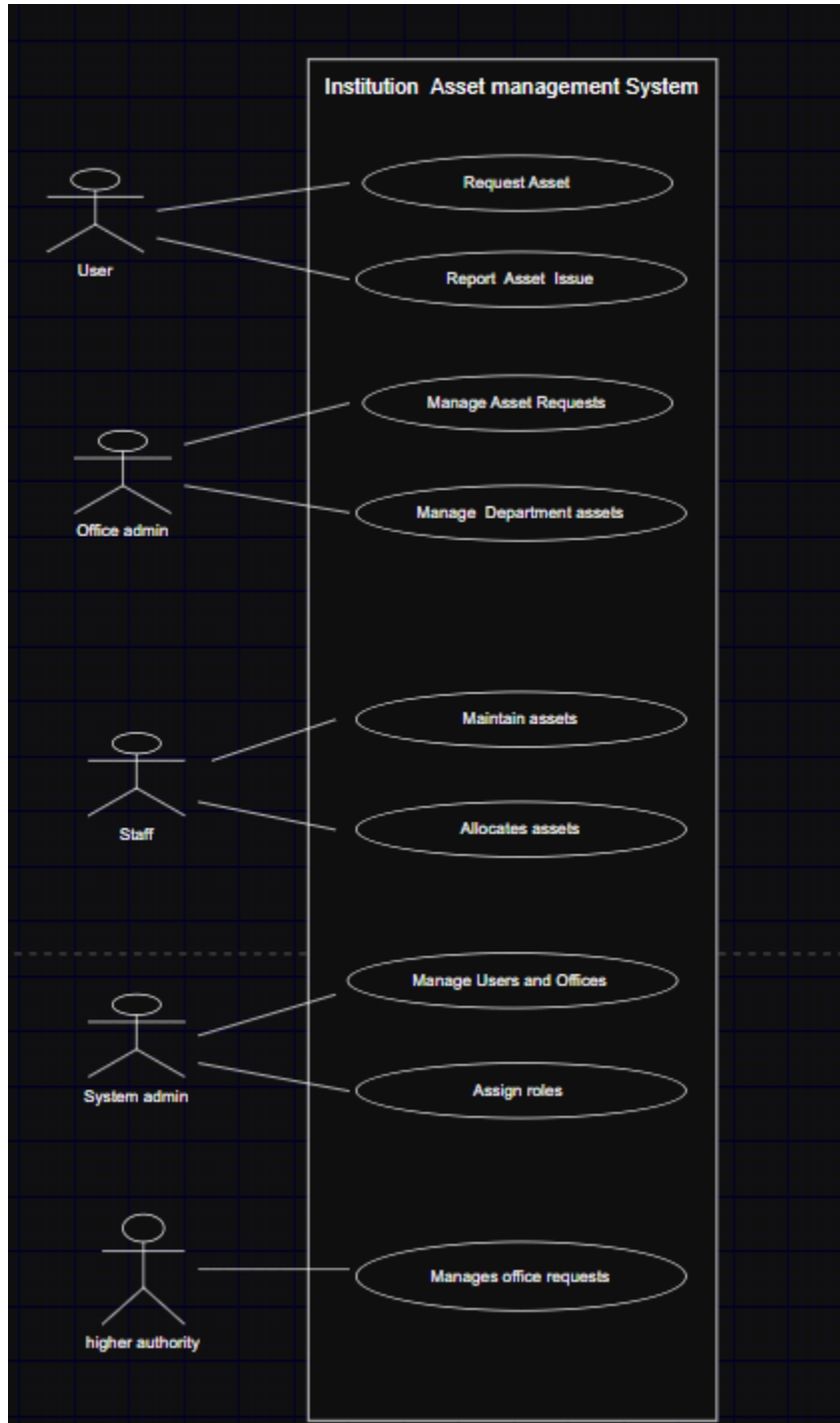
Usability: The system shall be user-friendly and require minimal training.

Scalability: The system shall support future expansion of users and assets.

6. System Models

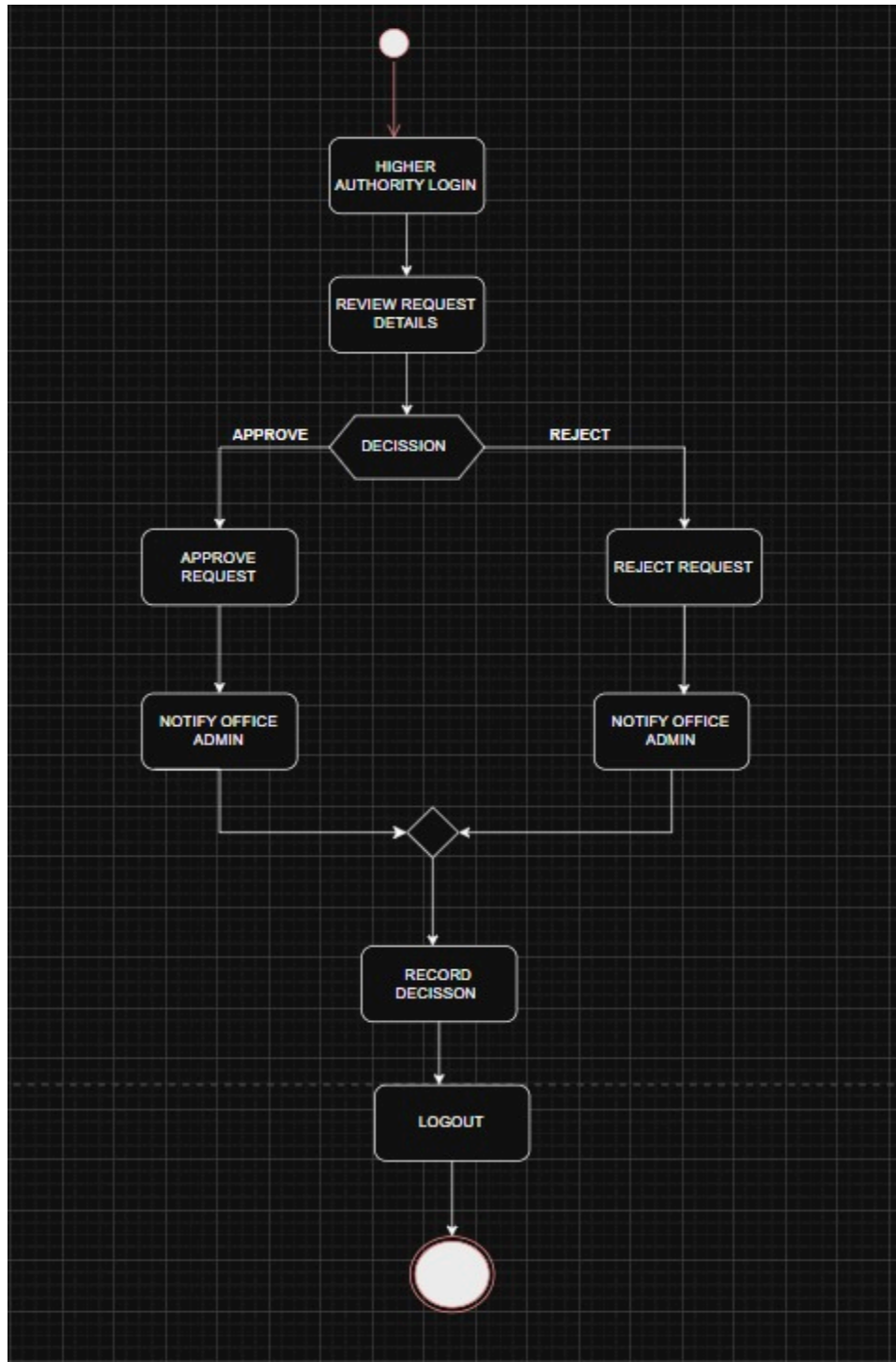
The following system models may be used:

Use Case Diagram

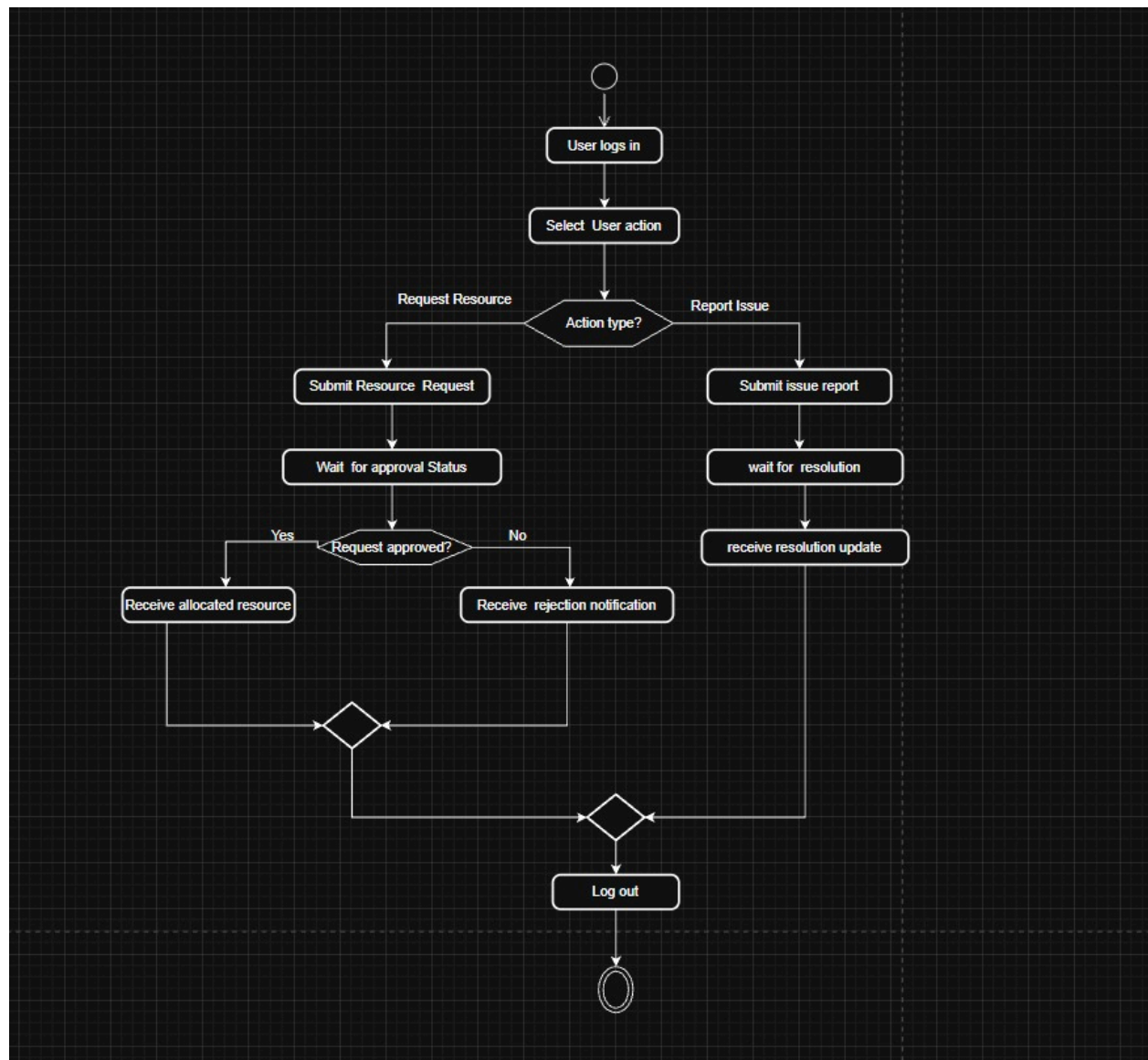


Activity Diagrams

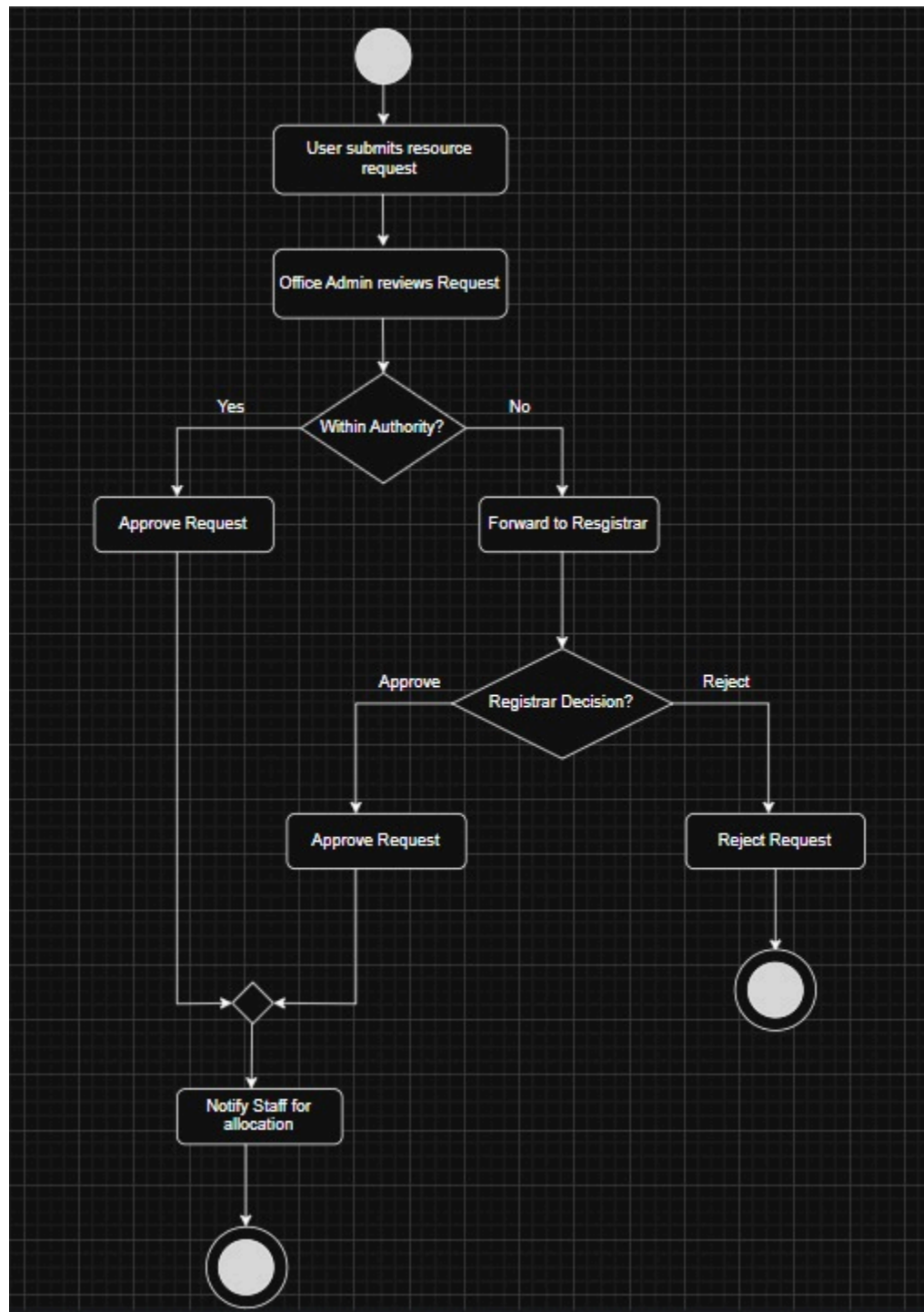
Higher authority asset request approval activity diagram



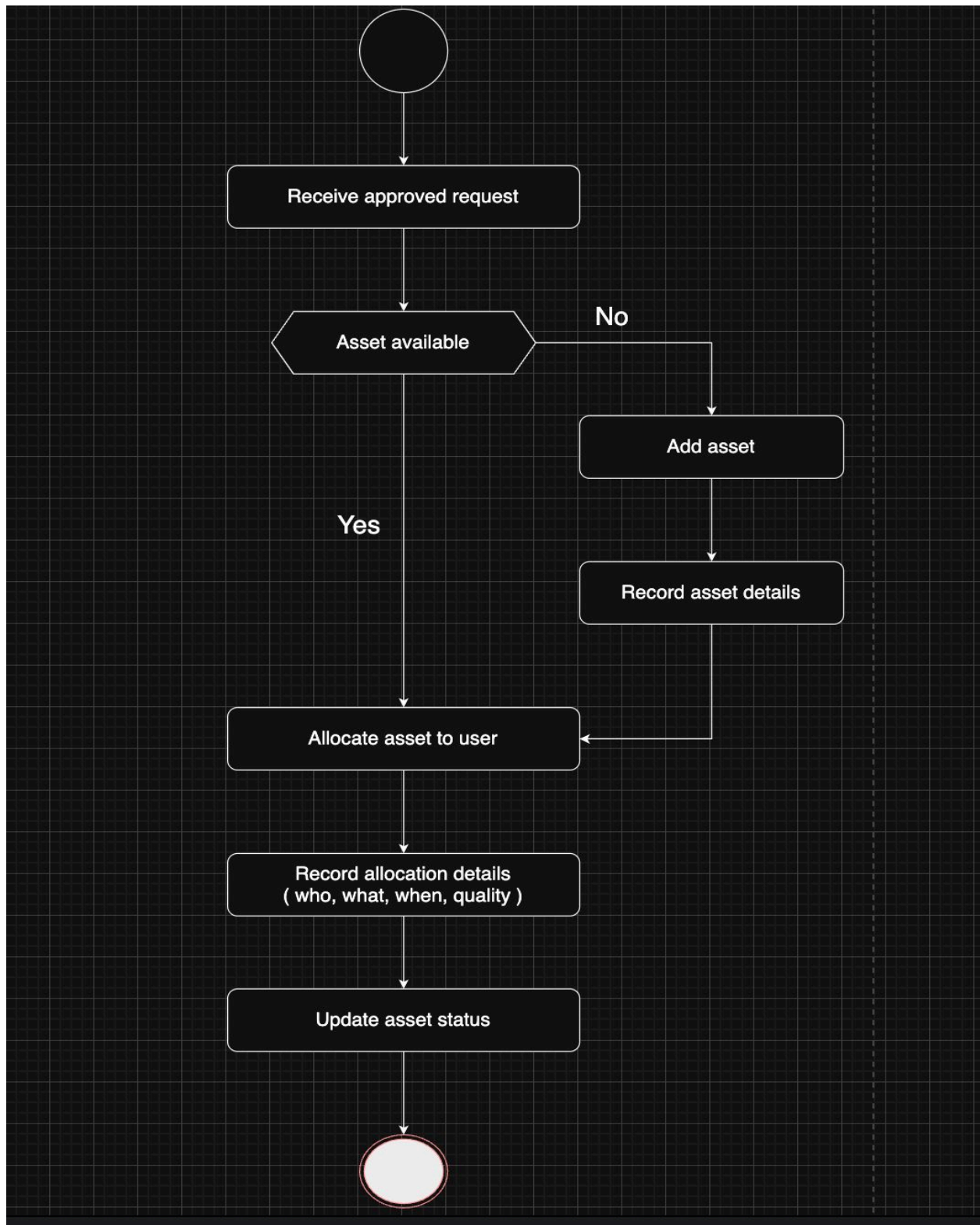
User activity (request/report) asset diagram



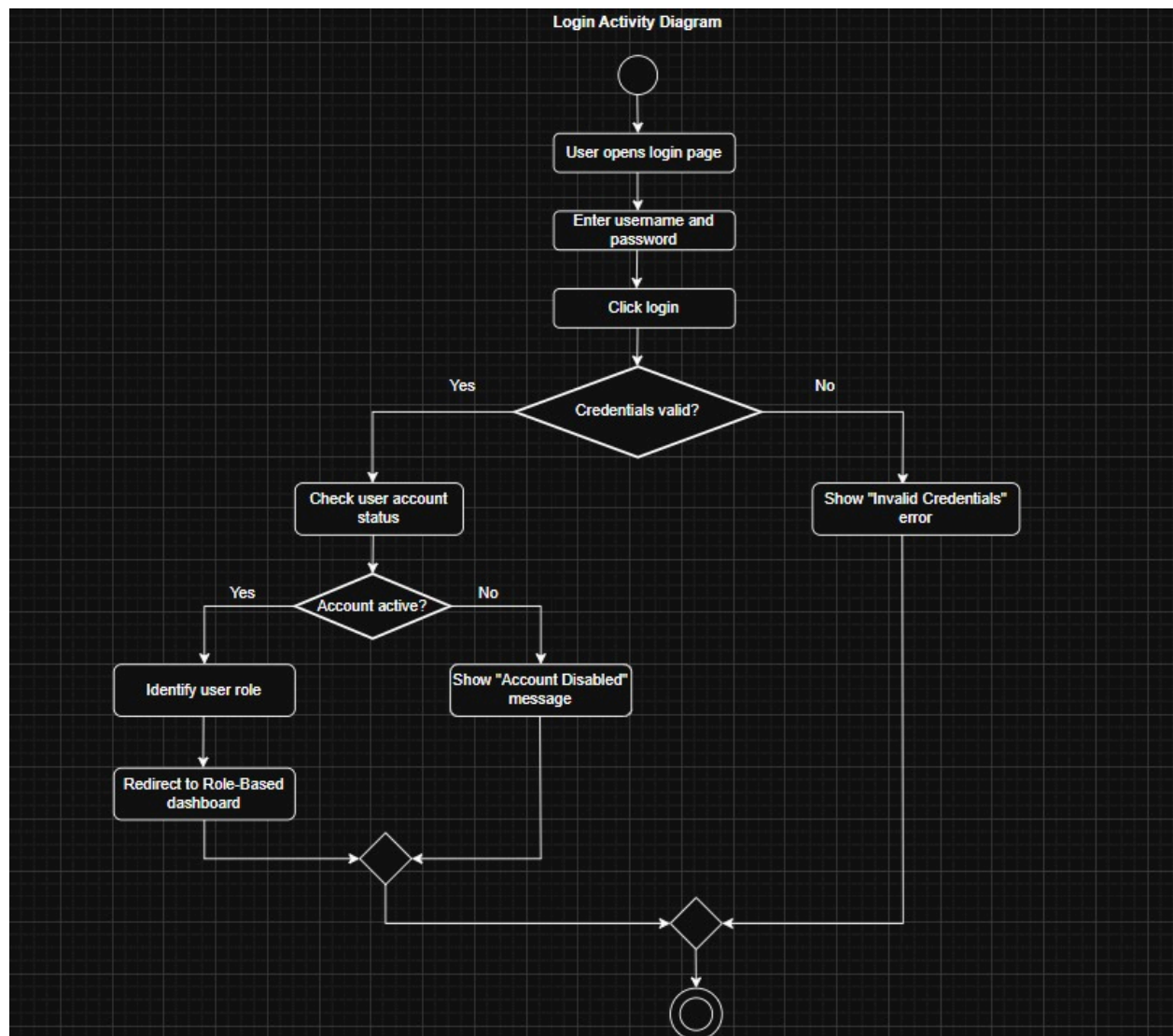
Office Admin asset request approval activity diagram



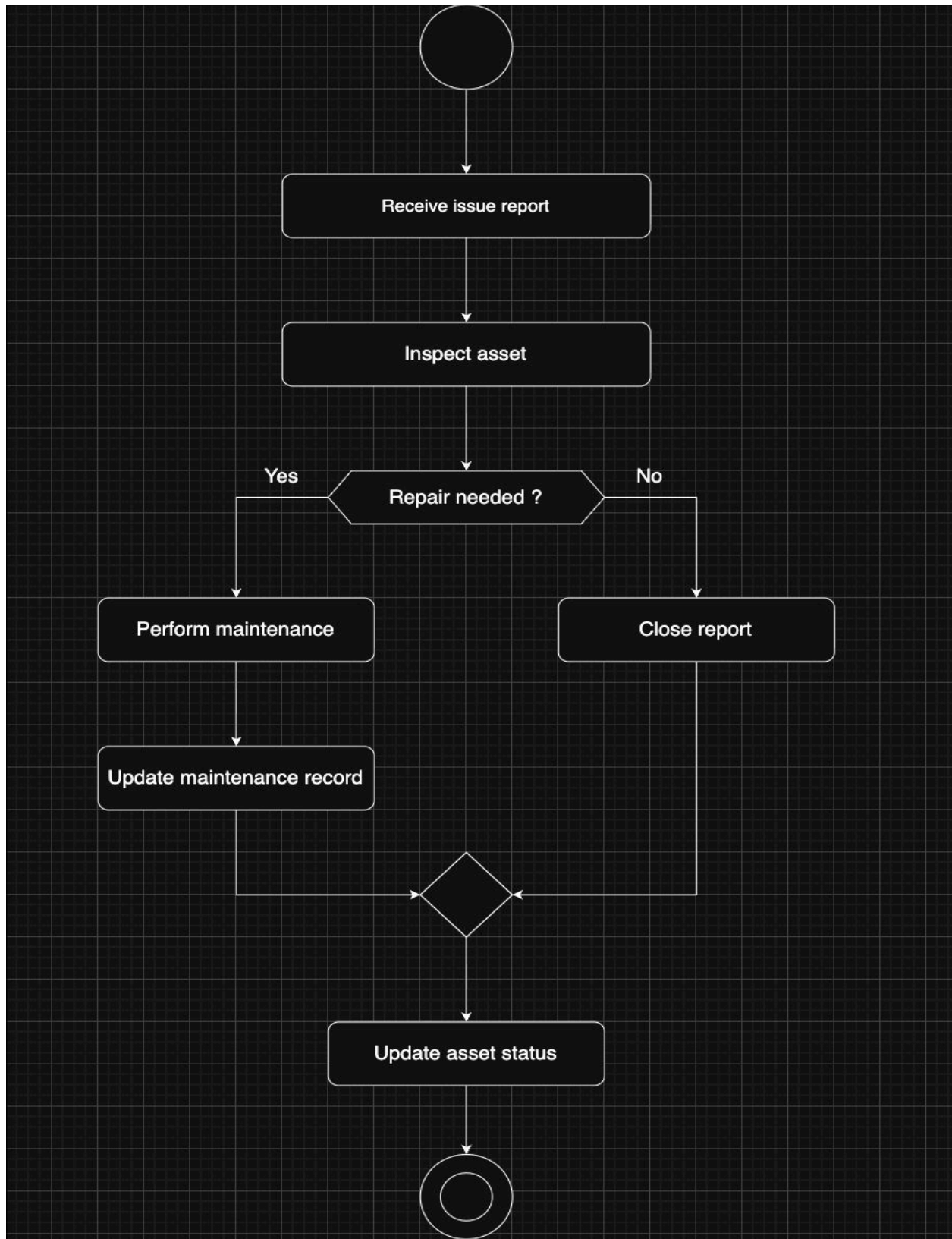
Staff asset allocation activity diagram



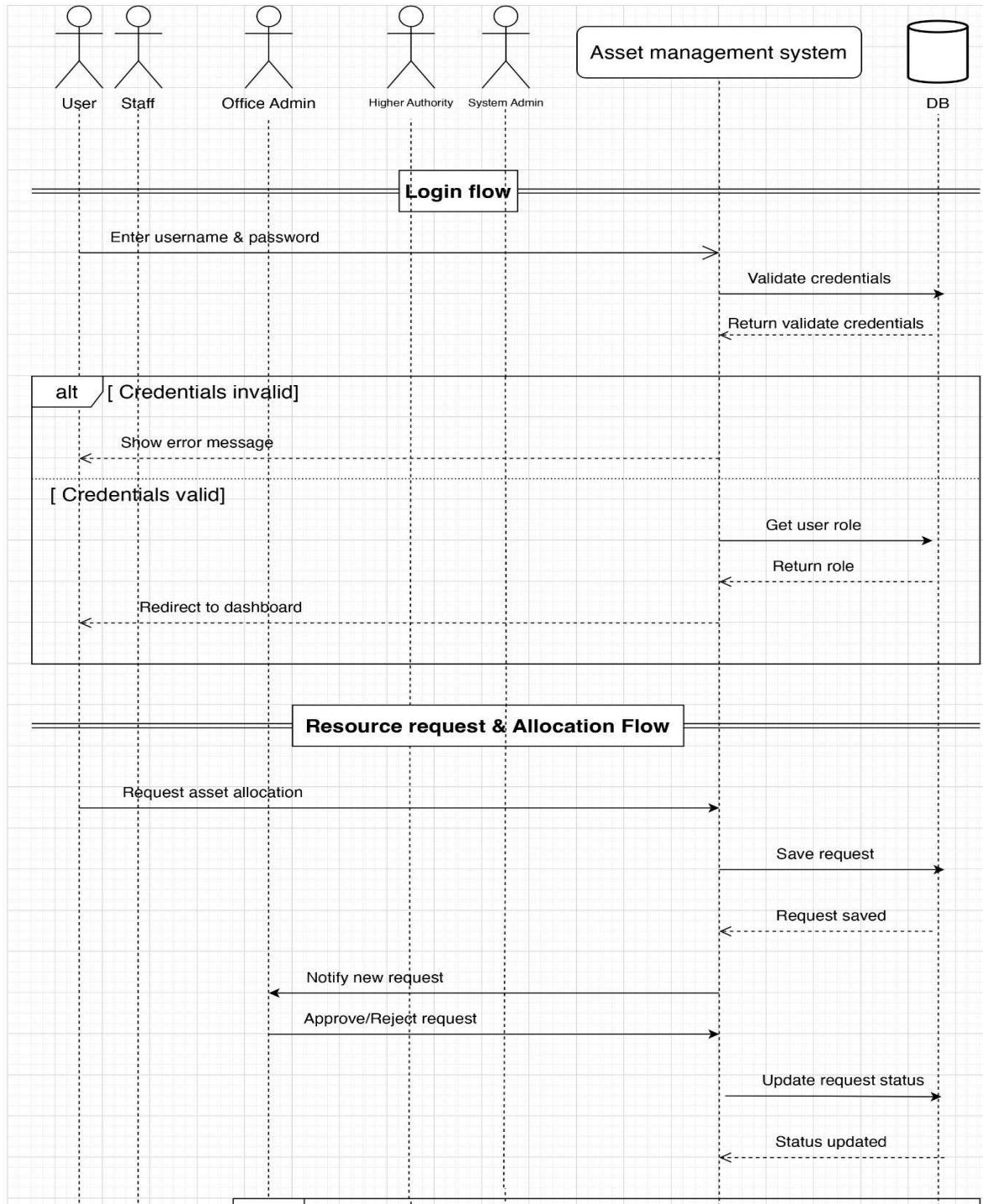
Login activity diagram

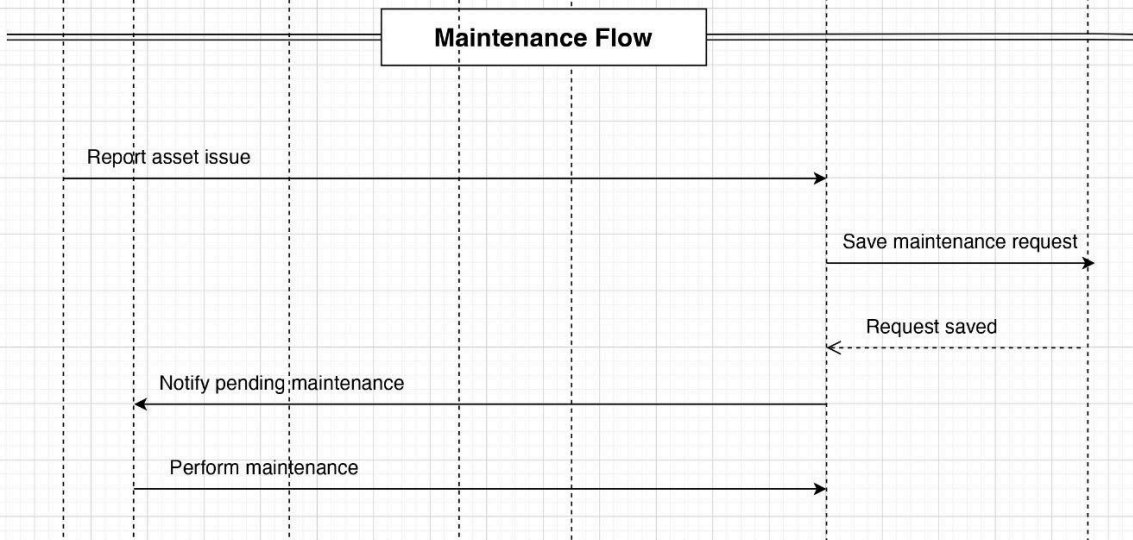
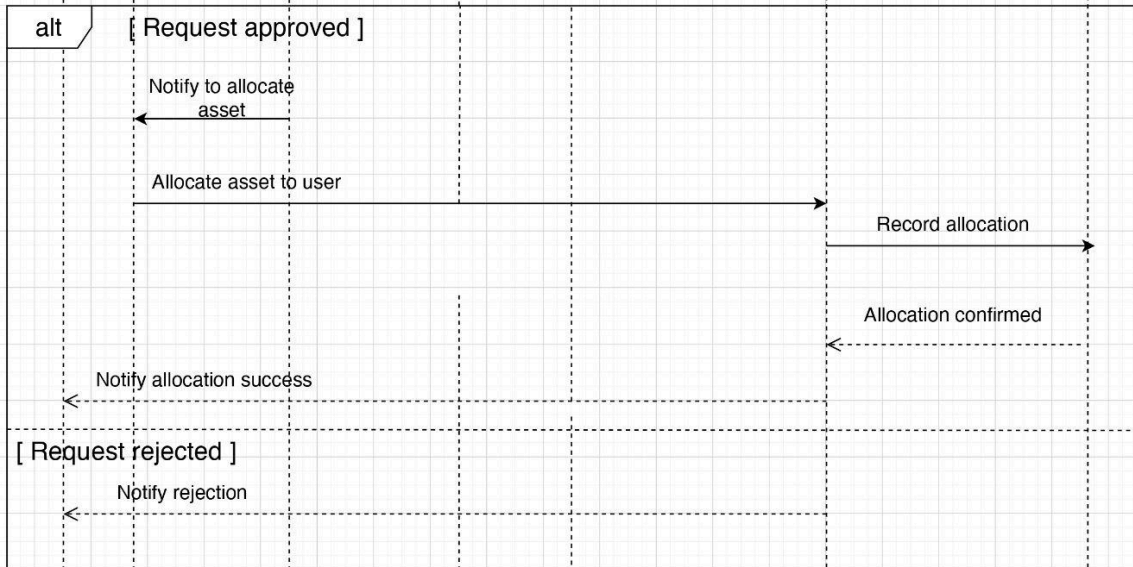
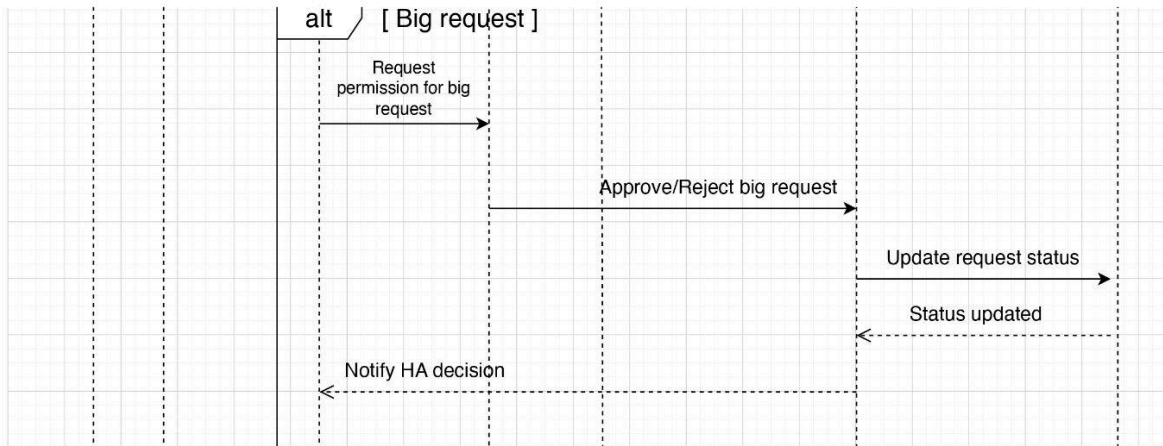


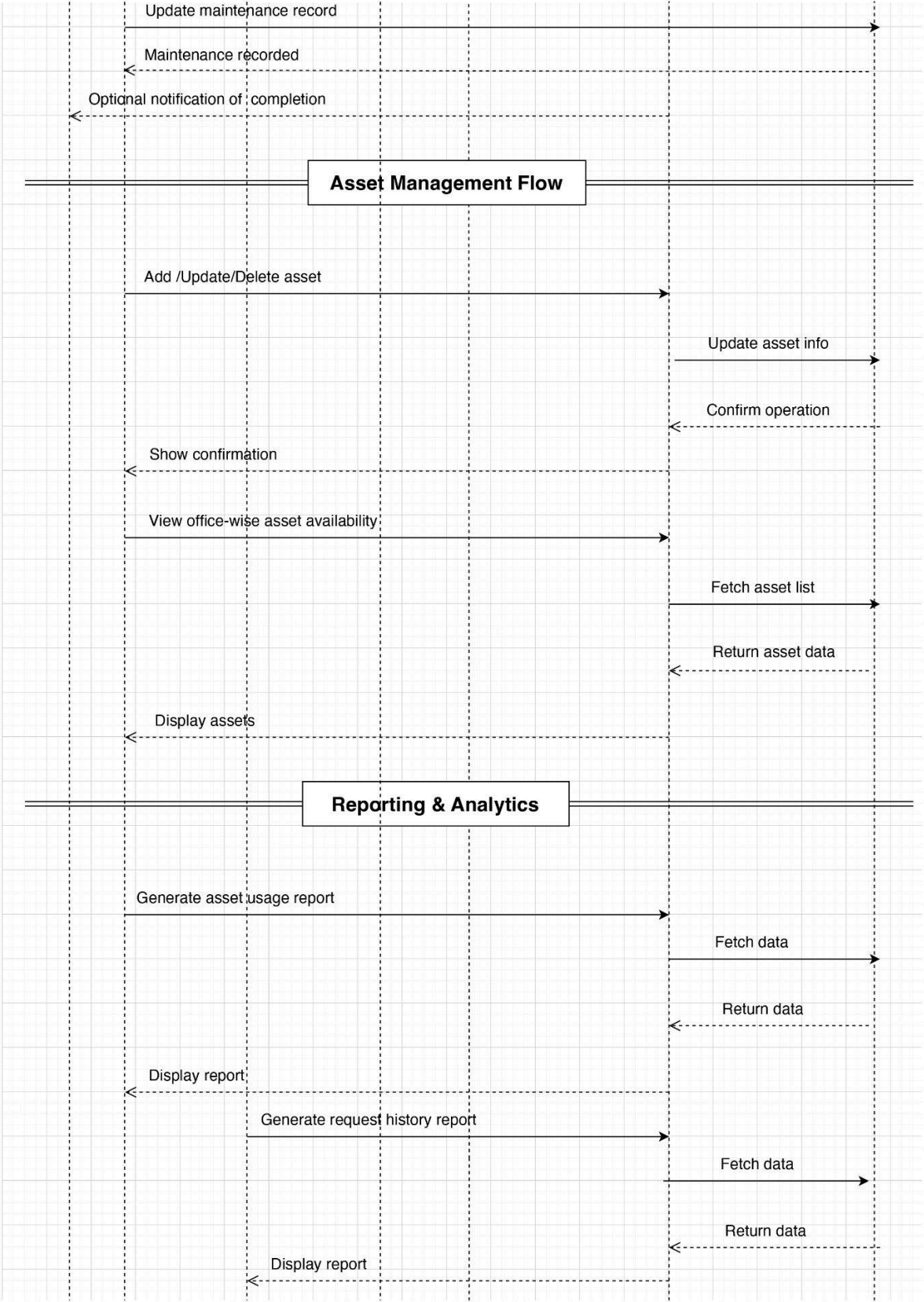
Maintenance activity diagram

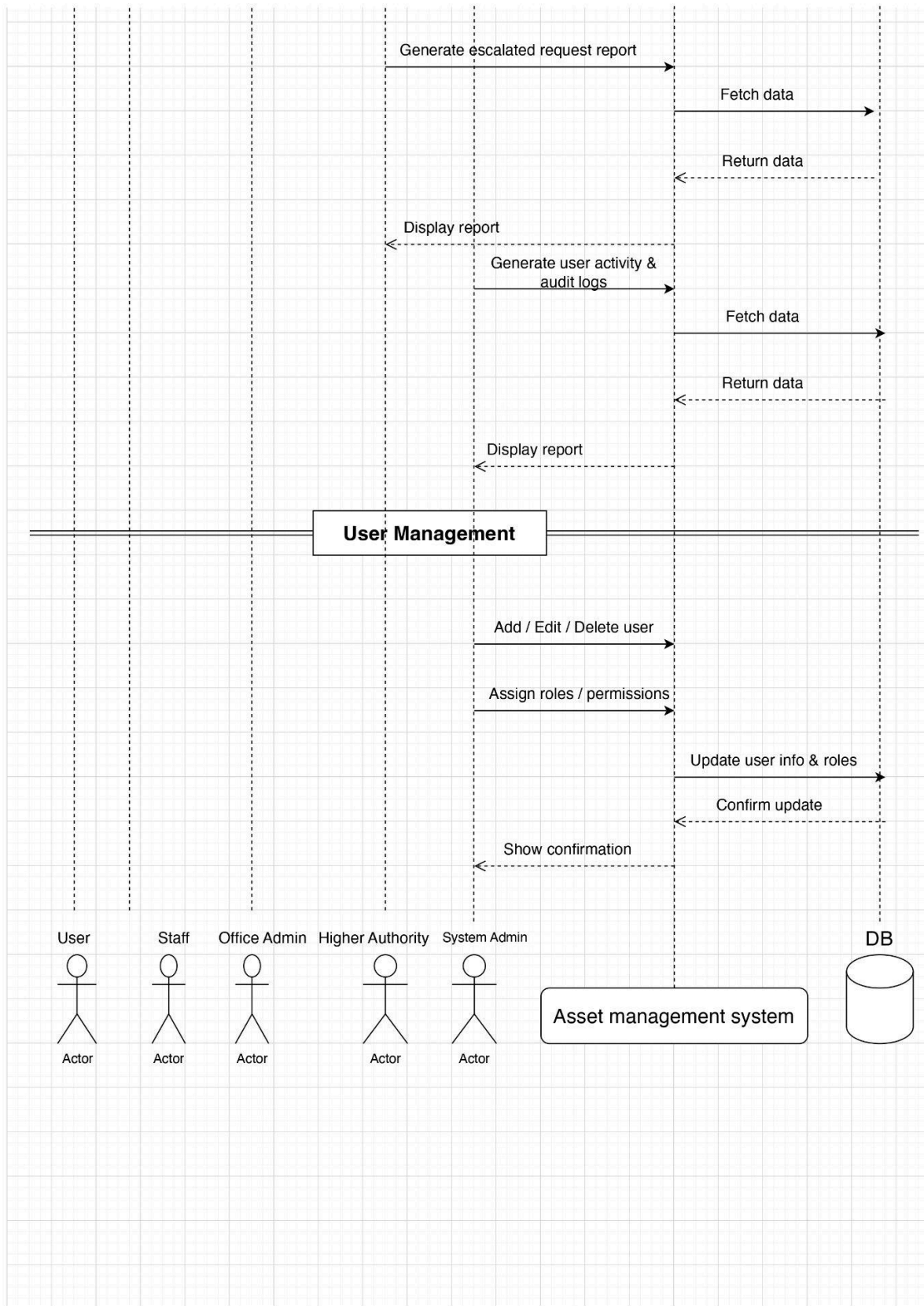


Sequence diagram









Entity Relationship (ER) Diagrams

7. System Evolution

The system is designed to support future enhancements such as mobile access, advanced analytics, and integration with external systems.

8. Appendices

8.1 Hardware Requirements

Server with minimum 8 GB RAM

Client systems with modern web browsers

8.2 Software Requirements

Web browser (Chrome, Firefox, Edge)

Backend server and database system

9. Index

Actors

Functional Requirements

Non-Functional Requirements

System Models